

Team Boundary Spanning: A Multilevel Review of Past Research and Proposals for the Future

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In response to increased complexity of work tasks, flatter organizational structures, and changing environmental conditions, organizational work teams must increasingly coordinate efforts across their boundaries and actively manage key relationships external to the team itself. Despite evidence of the importance of these processes—referred to as team boundary spanning—for both team and organizational success, significant gaps exist in our understanding of the nature of team boundary spanning, how and when these behaviors are carried out by teams, and the resulting impacts of team boundary spanning beyond that of enhanced team performance. Therefore, this article seeks to advance knowledge in this area by offering a taxonomy of team boundary spanning actions, reviewing the existing stream of team boundary spanning literature across multiple levels of analysis, and integrating this body of work with findings and perspectives from other boundary spanning research areas so as to stimulate fruitful avenues for future research.

Keywords: *boundary spanning; teams; group processes*

Challenges facing today's work organizations, including increased task complexity associated with knowledge work, changing economic conditions, global competition, and flatter work structures (Mohrman, Cohen, & Mohrman, 1995) are highlighting the need for organizational teams to coordinate interdependent work efforts and bridge disconnected parties by actively managing relationships external to the team itself. Commonly referred to as *team boundary spanning* or *team boundary management*, a

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team's efforts to establish and manage external linkages can occur within an organization (e.g., across marketing and manufacturing teams) or across organizational boundaries (e.g., to external customers, suppliers; see Ancona, 1990). Specific team boundary spanning behaviors may include representing the team to stakeholders, coordinating task activities with other groups, and seeking information from outside experts (Ancona & Caldwell, 1992a) and have been shown to be critical drivers to team performance (e.g., Gladstein, 1984) and organizational innovation (e.g., Hargadon, 1998). Recent perspectives on team functioning thus strongly emphasize deeper examination of externally directed team processes (e.g., Ancona, 1990; Ancona & Caldwell, 1992a; Faraj & Yan, 2009; Gladstein, 1984; Marrone, Tesluk, & Carson, 2007; Mathieu, Marks, & Zaccaro, 2001; Tesluk & Mathieu, 1999) in addition to the internal team processes such as conflict resolution and task coordination heavily studied in past research (see Mathieu, Maynard, Rapp, & Gilson, 2008).

While a great deal of empirical work in this area has been emerging over the past two decades, and some theoretical attention has been given to advancing team boundary spanning research (Ancona & Caldwell, 1998; Choi, 2002; Joshi, Pandey, & Han, 2009), significant gaps still exist in our understanding of the team boundary spanning phenomenon. First, conceptual contributions thus far have focused on identifying and exploring antecedents to team boundary spanning behaviors (e.g., Ancona & Caldwell, 1998; Joshi et al., 2009). Although contributing significantly to our understanding of factors that may facilitate these important behaviors, additional attention is needed toward learning more about the nature of team boundary spanning, how and when these behaviors are carried out by teams, and the resulting impacts beyond that of team performance. For instance, because teams play significant roles in coordinating efforts and knowledge across the organizational functions needed for organizational learning and innovation (e.g., Hargadon, 1998), and yet simultaneously face strong challenges in maintaining effectual internal processes required for their own sustainability (Ancona, 1990; Marrone et al., 2007), understanding the mechanisms through which teams effectively fulfill external demands while also preserving productive internal dynamics is important for maximizing both team and organizational effectiveness. Second, earlier theoretical contributions (e.g., Choi, 2002) could not include the empirical findings that have emerged in recent years and, thus, were unable to comprehensively assimilate Ancona and colleagues' initial work on team boundary spanning with subsequent findings across different organizational settings and team types. Finally, integrations of the team boundary spanning literature with closely related literature, such as organizational boundary spanning, have not yet been fully explored and provide rich avenues for advancing team boundary spanning research.

To address these gaps, this article comprehensively reviews the literature on team boundary spanning, focusing directly on seminal work advanced by Ancona and subsequent contributions to this stream, and integrates this body of work with findings and perspectives from related disciplines in ways that stimulate fruitful future research directions. A comprehensive definition and taxonomy of team boundary spanning actions are offered first, and the associated outcomes for the team, interdependent entities that are mutually reliant on the

boundary spanning team, and embedding organizations are delineated. Further, given that team boundary spanning actions originate from the behaviors and actions of its individual members, and because team boundary spanning actions reflect and are embedded in the team's linkages to external actors, the conceptualization and review of team boundary spanning consider these multiple levels of analysis: member, team, and network (behavior internal to a network of multiple teams or actors). This review contributes, first, by offering a deeper understanding of the nature and functionality of the team boundary spanning construct through an emphasis on critical parameters, processes, and interrelationships within and across levels. Second, it contributes by achieving a more thorough and in-depth understanding of the state of team boundary spanning research, highlighting key themes and conclusions across studies, including for instance emerging insights into relationships among external team activity, internal team functioning, and team effectiveness. And finally, through integrations with other related research, this article advances future research directions within the team boundary spanning arena, as well as in related research areas, in new ways.

A review and integration of this scope also contributes to our understanding of emerging organizational problems informed by team boundary spanning. Uncertain and unpredictable environments, including strong competitive pressures, require continuous innovation, adaptability, and the development of high-quality products and services with increasingly short lead times for survival (Denison, Hart, & Kahn, 1996; Mohrman et al., 1995). Through boundary spanning efforts, teams bridge otherwise diverse and disconnected parties and act as critical conduits for information transfer, knowledge creation, and innovation (e.g., Argote, McEvily, & Reagans, 2003; Hargadon, 1998). Further, as knowledge work becomes increasingly complex, specialized, and nonroutine, tasks are no longer easily performed independently or sequentially but require interdependent and coordinated action across various parts of the organization and even across organizations, as is the case of strategic alliances (DeChurch & Mathieu, 2009). Finally, as organizations move away from traditional structures, such as the bureaucracy, for flatter, team-based ones, they must rely on boundary spanning teams to coordinate work efforts, identify improvements to organization-wide processes, and make strategic decisions (e.g., Cross et al., 2000). As such, reviewing, integrating, and advancing research within this domain offers strong promise for helping organizations manage effectively and operate efficiently in today's business environment.

To select specific articles to be reviewed, I first conducted electronic searches for early work by Ancona (formerly Gladstein) and Caldwell. Next, using their research as a guide, I searched for keywords used by these scholars (e.g., *team boundary spanning*, *team boundary management*, *ambassadorial activity*) to identify subsequent contributions to this stream. I also scanned tables of content of major management journals and citation indexes. It is important to note that while several additional literature streams include within them considerations of team connections to other teams or actors—knowledge-sharing research or social network research are just two examples—the literature review that follows is intentional in that it focuses exclusively on the stream of team boundary spanning literature to allow for in-depth and comparable assessments of studies directly examining team boundary spanning activities and processes. I begin with a definition and description of team boundary spanning.

Definition of Team Boundary Spanning

Drawing upon the team boundary management literature as well as past research on organizational-level boundary spanning, *team boundary spanning* is defined as the team's actions to establish linkages and manage interactions with parties in the external environment (Ancona, 1990; Ancona & Caldwell, 1992a; Marrone et al., 2007). Such actions can assist the team, and others linked to the team, in meeting performance goals and task objectives. Example actions include representing the team to outsiders (e.g., updating and seeking feedback from upper management on team progress), searching for information (e.g., contacting subject matter experts for project-related expertise), and coordinating task performance with other external groups (e.g., communicating action plans and delivery deadlines with other interdependent teams; see Ancona & Caldwell, 1992a; Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). Here, *external environment* refers to the team's embedding environment (that which resides outside of the team's own boundary) and may include actors or other teams residing within or outside of the boundary spanning team's host organization. Thus, boundary spanning actions may extend beyond a single organization, serving to link the team (and its organization) with other organizations (e.g., Mathieu et al., 2001 discuss the boundary spanning efforts involved in emergency response across four mutually interdependent teams and two organizations: county government and hospital).

The benefits of team boundary spanning can similarly extend beyond the focal team itself to the performance of other parties that are interdependent with the boundary spanning team and to the achievement of higher order organizational and cross-organizational goals (patient survival in emergence response; Mathieu et al., 2001). Further, although team boundary spanning behaviors are consistent in their focus toward achieving task-related goals and performance objectives (Ancona & Caldwell, 1992a), these actions may or may not result in desirable consequences. While a team may diligently seek information from outsiders, the information received may be inaccurate or incomplete, causing disruptions in team performance. Or a team may prolong scouting for technical information, diverting resources and undermining efficiency and innovation (see Ancona & Caldwell, 1992a). Therefore, I note that the definition of team boundary spanning is *not* restricted to include only those behaviors that are consequently beneficial or functional.

It is important to place the concept of team boundary management within the broader stream of teams research (see Mathieu et al., 2008) as well as compare these activities with internal team processes heavily studied in past research. Team boundary spanning activities are referred to as *external* team processes, which reflects the nature of the team's interactions with parties external to the team itself, and therefore, they differ in a meaningful way from *internal* team processes, which are intrateam and involve the nature of team member interactions with one another to internally develop strategies, coordinate workloads, and manage interpersonal conflict (Marks, Mathieu, & Zaccaro, 2001). Both internal and external team processes share commonalities in that they represent team-level actions carried out for the purposes of meeting collective performance goals (Marks et al., 2001; Mathieu et al., 2008) and originate in the behaviors and actions of individual team members (Kozlowski & Klein, 1999). External team processes (i.e., team boundary spanning) are distinct, however, in that they capture the interactions *across* the team to parties in the embedding environment such

as clients, customers, industry experts, and other mutually interdependent teams (rather than capturing the interactions across the team members as is the case in internal team processes). Consequently, analogous to team members nested within their team, teams that boundary span are additionally nested within their larger set of relationships (be these dyadic, triadic, or greater).

An explicit consideration of multiple levels of analysis is therefore critical to the conceptualization of team boundary spanning. First, the team boundary spanning behavior of a single, focal team is a team-level construct, one that originates from the individual behaviors of team members who engage in the actual boundary spanning work on behalf of their team. Because the individual behavioral contributions to team boundary spanning may or may not be isomorphic or converge among members (Kozlowski & Klein, 1999), and instead may vary in amount and type depending on task and team interdependencies (e.g., a focal team member or a select few members may carry out the majority of the team's boundary spanning function), this team-level construct may be best described as a *process composition model* (Chan, 1998). Process composition models assume that the basic process (interacting with parties outside the team boundary) is homologous across levels, and thus, there is no simple algorithm (rwg, dispersion statistic, ICC, etc.) that describes precisely how lower level processes compose higher level ones. Rather, as Chan notes, the process is first specified at the lower level and then the analogue parameters and interrelationships are described at the higher level. In the case of composing the team boundary spanning construct, the processes through which members perceive, evaluate, and interact with their team's embedding environment are assumed analogous to the processes by which teams gain shared understandings of their environment and establish external linkages.

In addition, because team boundary spanning reflects and is embedded within the team's linkages to other external actors, the conceptualization of team boundary spanning is not confined to the team level. From a social network perspective, team boundary spanning can be represented—both visually and conceptually—as the relational ties established between various actors (whereby actors are equivalent to single teams and/or other social entities). Consequently, attributes and behaviors of an individual actor (e.g., a focal team's boundary spanning) and the structured pattern of relationships it is embedded within (i.e., the network characteristics and behavior) are relevant (Wasserman & Faust, 1994). Therefore, as advanced in the multiteam systems (MTS) literature, which highlights the coordination efforts across two or more mutually interdependent teams that share at least one common goal, team boundary spanning is also appropriately conceptualized at a higher level of analysis (labeled as *system-level* and reflective of *interteam* coordination; DeChurch & Marks, 2006). At this level, *internal* and *within* represent the aggregated interactions across the component teams comprising the system. In the text that follows, when referring to this higher level of analysis, I use a more generic term, *network boundary spanning*, to depict boundary spanning actions occurring not only within a system of mutually interdependent teams but also within a set of diverse actors with varying degrees of interdependence, such as multiple R&D teams sharing information in a large pharmaceutical organization or a public accounting team, its client, and an in-house expert consulted for technical expertise.

In summary, team boundary spanning encompasses the various activities through which teams actively manage interactions with their embedding external environment and complete their work. These *external* team processes enable the team to attend to its external

environment and facilitate performance and are in addition to internal team processes traditionally studied in past teams research. Because boundary spanning actions are embedded in the team's linkages to other external actors and because such actions originate from the behaviors and actions of its individual members, the conceptualization of boundary spanning considers these multiple levels of analysis. These externally directed activities allow for diversity in the information obtained by the boundary spanning team (Hansen, 1999), opportunities to negotiate project expectations and requirements in ways that facilitate team performance (Ancona, 1990), and coordination across mutually interdependent teams critical for organizational learning, innovation, and network performance (DeChurch & Mathieu, 2009; Mohrman et al., 1995).

Taxonomy of Team Boundary Spanning Actions

Before moving to a deeper discussion of relationships among antecedents, behaviors, and consequences, my review of the literature begins with offering a taxonomy of team boundary spanning actions that draws upon past research and organizes extant findings across three broad behavioral categories—*representation*, *coordination of task performance*, and *general information search*. Table 1 presents the taxonomy, delineating each category across four dimensions: who (nature of the target), what (primary functions), how (level of interdependence), and why (expected outcomes). The taxonomy aligns closely with the emergent findings from seminal work (e.g., Ancona & Caldwell, 1992a) that focused primarily on behaviors and outcomes for the boundary spanning team and its members, as well as later advances (e.g., Marks et al., 2001) that have directly advanced our understanding of boundary spanning actions within systems of teams and associated outcomes at higher levels. Thus, the categories are relevant to members, teams, and networks of multiple teams or actors.

The *representation* category reflects team actions that persuade other parties of team decisions, ask for resources, and protect the group. Whereas representation behaviors can certainly occur at all levels within (and across) organizations, the target actors often hold greater power than the boundary spanning team (see Ancona & Caldwell, 1992a), such as upper management or external clients who can additionally reward or sanction teams based on performance. Example actions include soliciting upper management feedback on team progress, termed *ambassadorial* by Ancona and Caldwell, and serve several primary functions such as clarifying expectations, forming impressions, and advocating for the team. Because interactions between the boundary spanning team and the target entity are moderately (albeit not equally) interdependent, representation behaviors produce outcomes affecting all parties involved, including the team (e.g., receipt of higher level support or commitment to team decisions) and the target (e.g., being kept abreast of team progress enables real-time assessments of organizational initiatives and should allow for higher quality planning and resource allocation). Benefits also arise for the boundary spanning team's embedding organization, as the actions to proactively clarify and meet external client expectations should reflect positively upon both the team and the organization it is representing (e.g., Bettencourt, Brown, & MacKenzie, 2005).

Coordination of task performance represents team actions that coordinate work activities with mutually interdependent entities to accomplish individually and jointly set task goals.

Table 1
Taxonomy of Boundary Spanning Actions

Behavioral category	Description	Dimension 1: Nature of target	Dimension 2: Primary functions/ objectives	Dimension 3: Level of coupling/ interdependence	Dimension 4: Representative outcomes
1. Representation	Actions that persuade others of team decisions, request resources, and protect the group	To those of greater power (e.g., upper management, clients)	Set and clarify expectations, form impressions, advocate for team needs	Medium	<i>For team:</i> upper-level support, accurate diagnosis of target needs, enhanced team reputation <i>For target:</i> clarity of expectations (team and target), quality resource monitoring <i>For organization(s):</i> improved strategic decision making, enhanced reputation
2. Coordination of task performance	Actions that coordinate work activities with interdependent entities to accomplish task goals	To those of equal power (e.g., other interdependent teams)	Synchronize work efforts, inputs, and outputs of interdependent entities; monitor strategy; and progress toward joint goal achievement	High	<i>For team and target:</i> lateral support and cooperation, shared awareness of dependencies, meeting deadlines, team efficiency, quality planning, coordination, and communication across entities <i>For organization(s):</i> synchronization of efforts, organization learning, adaptation, efficiency in operations, and achievement of organization/cross-organization goals
3. General information search	Actions that access outside parties for general or technical information or expertise	To those with specific knowledge (e.g., industry experts)	Gain expertise (problem or project specific) and understand general environment	Low	<i>For team:</i> receipt of informational resources and expertise, shared awareness of environmental opportunities/threats, team learning and innovation <i>For target:</i> potential future reciprocity gains <i>For organization(s):</i> innovation

These behaviors are often directed laterally and involve external teams or actors tightly coupled with the boundary spanning team. Ancona and Caldwell (1992a) describe a team's efforts to negotiate delivery deadlines with other teams. Further, Marks et al. (2001) introduce inter-team process, or the joint efforts across interdependent teams to monitor cross-team progress and work flow (during action phases) as well as collectively develop strategy, plan, and evaluate (during transition phases). Given the high degree of interdependence, this category of boundary spanning actions provides a variety of outcomes equally relevant to the focal team and the other interdependent entities, such as the achievement of collectively held goals, lateral support and cooperation, efficiency, and so forth. Further, given the strong lateral linking function essential for combining diverse ideas and identifying improvements in organization-wide processes, enhanced organizational innovation, efficiency, and adaptability are also expected (Mohrman et al., 1995).

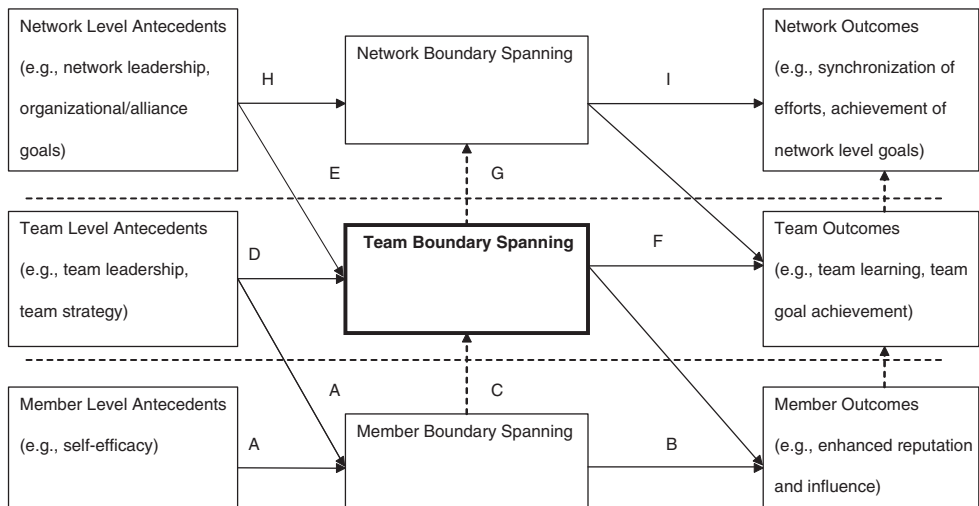
Finally, *general information search* reflects team actions to parties that hold specific knowledge and for gaining access to information and expertise. These behaviors enable teams to gain problem- or project-specific expertise as well as to understand the general environment they operate within, such as trends, opportunities, and threats. Target actors are often (although not necessarily) loosely coupled with the focal team—for example, contacting a former mentor for technical advice—that share little if any mutual dependency with the focal boundary spanning team. Thus, this category of boundary spanning actions generates outcomes primarily relevant to the boundary spanning team itself and its embedding organization, such as shared understandings of environmental attributes, and team learning and innovation, the latter of which has been linked to organization innovation (Hargadon, 1998).

Most work to date in the team boundary spanning literature has focused on behaviors residing in the representation and coordination categories (most often focused on ambassadorial and task coordination from Ancona & Caldwell, 1992a; Joshi et al., 2009). There have been few attempts to differentiate antecedents or consequences across categories, despite the proposed differences noted above.

Multilevel Model of Team Boundary Spanning

I move next to the multilevel model presented in Figure 1 that explicitly considers the nesting of member, team, and network of teams or actors inherent to the team boundary spanning construct. Through the use of this model, I organize, examine, and discuss key findings stemming from major works within the team boundary spanning literature. I start first by reviewing work that has considered the facilitators and outcomes of boundary spanning actions undertaken by members on behalf of the team. Next, extant findings concerning antecedents, mediators, moderators, and outcomes at the team level follow and provide for the majority of the review. Last, given the nesting of teams in organizations and, more proximally, within their network of linkages to other external actors, I conclude with a discussion of extant work considering of antecedents and consequences at the network level.¹ As relationships are reviewed in the text below, they are cross-referenced to labels in the figure so as to visually denote the appropriate path. Dashed lines in the figure represent

Figure 1
Proposed multilevel model of member, team, and network boundary spanning



Note: The dashed lines denote compositional processes through which lower level phenomenon are compiled to form higher level phenomenon. For simplicity, moderations and mediating mechanisms are not shown.

bottom-up processes through which lower level boundary spanning processes emerge to compose higher level phenomenon, whereas solid lines reflect directional relationships among constructs within and across levels. Finally, critical functions (e.g., representation, coordination of task performance, and general information search) and interrelationships among parameters are assumed homologous across levels (e.g., task-appropriate strategies likely facilitate boundary spanning at all three levels), yet unique antecedent or outcome variables may also arise when moving up levels (Kozlowski & Klein, 1999). A summary of major empirical works is provided in Table 2.

Team Member Boundary Spanning Behavior

Antecedents (Paths A). Although boundary spanning is critical for team performance, it is also highly challenging and taxing for individuals engaged in these efforts on behalf of their teams. These challenges arise because boundary spanning individuals in teams continually balance attention between internal and external demands (Choi, 2002) and face conflicting expectations across internal and external constituencies (Katz & Kahn, 1978). Consequently, factors relating to motivation, competency, and task-appropriate strategies should be critical for increasing member engagement, persistence, and effectiveness in boundary spanning behavior. Existing findings have lent support for these

Table 2
Exemplar Empirical Studies of Team Boundary Spanning

Study	Team type and number	Method	Boundary spanning variables studied	Other variables studied	Key findings
Ancona & Caldwell (1988)	38 new product development teams	Interviews, logs	Scout Ambassador Sentry Guard (f)	Antecedents of boundary activities (emerged via inductive theory building)	Groups use specific strategies for managing their environment, each associated with a set of activities. Activities appear to be predicted by individual, group, and organizational characteristics.
Ancona & Caldwell (1990)	45 new product development teams	Interviews, surveys, archival data	Ambassadorial Task coordination Scouting Guarding (a)	Task environment characteristics, individual characteristics	Task environment characteristics (only resource availability, stage of product), individual characteristics (leadership, tenure, time spent on project, diversity in experience), and organization predicted boundary spanning.
Ancona & Caldwell (1992a)	45 new product development teams	Interviews, surveys, logs, observation	Ambassadorial Task coordination Scouting (a)	Internal processes, cohesiveness, team performance (g) (budget, efficiency, operation, innovation), team ratings of performance	Teams developed distinct strategies toward outsiders. The pattern of external behaviors is a better predictor of performance than frequency of communication is. Teams with comprehensive strategies were the highest performers.
Ancona & Caldwell (1992b)	45 new product development teams	Interviews, surveys	Communication with outsiders (c)	Team functional and tenure diversity, team performance (g) (budgets, innovation), team ratings of performance	Functional diversity predicted external communication, which predicted innovation. Tenure diversity predicted internal processes, which positively predicted team rated performance.
Ancona (1990)	5 consulting teams	Interviews, surveys, logs, observation	Leader strategies: Informing Parading Probing (f)	Internal processes, team performance (emerged via inductive theory building)	Team leader strategies toward the environment varied across teams. Probing teams (highest external orientation) were rated as the highest performers but suffered poor satisfaction and cohesion in the short term.

(continued)

Table 2 (continued)

Study	Team type and number	Method	Boundary spanning variables studied	Other variables studied	Key findings
Cross, Yan, & Louis (2008)	Single organizational case study of banking teams	Interviews, surveys, archival, observation	Boundary spanning Boundary buffering Bringing up boundaries (f)	Not applicable	Organizational transformation from functional to a team-based structure resulted in boundary work (a) migrating to lower levels of the organization and (b) increasing in frequency and significance.
Druskat & Kayes (1999)	21 production teams and 11 maintenance teams	Interviews, surveys	Organizational Awareness Resourcefulness Relationship building (b)	Contextual variables, process variables (interpersonal behaviors, performance strategies)	Contextual antecedents (reward systems, autonomy/influence, external leadership) affect different teams differently such that team internal differentiation and external integration moderated the relationship between context and process.
Druskat & Wheeler (2003)	66 external leaders of manufacturing teams	Interviews, survey	Leadership: Relating Scouting Persuading (f)	Team-focused behavior, leadership effectiveness (emerged via inductive theory building)	Superior leaders actively engaged in both boundary spanning behaviors and team- focused behaviors over time, shifting attention back and forth from the team to the organization.
Edmondson (1999)	51 work teams	Survey	Coordinating objectives, schedules, and resources across team boundaries (b)	Team psychological safety, clear direction, context support, team composition, team task, team performance (g)	Team psychological safety and context support positively predicted boundary spanning activities. Cross-functional teams had more boundary activity spanning than functional teams.
Edmondson (2003)	16 operating room teams	Interviews, observation	Boundary spanning with other organizational groups (f)	Antecedents and outcomes to team learning processes (emerged via inductive theory building)	Team leadership coaching predicts team boundary spanning. Team boundary spanning positively correlates with team learning and implementation success.

(continued)

Table 2 (continued)

Study	Team type and number	Method	Boundary spanning variables studied	Other variables studied	Key findings
DeChurch & Marks (2006)	64 multiteam systems (MTS) in a laboratory simulation	Manipulation (leader training), observations, surveys	Interteam (MTS) coordination (h)	Leadership training, leadership, intrateam coordination, team performance (i), MTS performance (i)	Leadership training manipulations improved MTS functional leadership, which directly affected interteam coordination. Interteam coordination fully mediated the relationship of MTS leadership and MTS performance.
Faraj & Yan (2009)	64 software development teams	Surveys	Boundary spanning Boundary buffering Boundary reinforcement (b)	Psychological safety, task uncertainty, resource scarcity, team performance (g) (goal achievement)	Boundary work predicted psychological safety and team performance. Relationships are moderated by task uncertainty and resource scarcity.
Gladstein (1984)	100 sales teams	Surveys, archival	Boundary management (e)	Leadership, structuring of activities, intragroup processes, task variables, team effectiveness (g), sales revenue	Psychological safety is not a mediator between boundary work and performance. Group members unexpectedly differentiated between internal and external components. Task variables did not moderate the process to performance relationships. Internal processes predicted member-rated satisfaction and team performance but not actual sales revenue.
Marks, DeChurch, Mathieu, Panzer, & Alonso (2005)	46 MTS in a laboratory simulation	Manipulation (goal hierarchy), observations, surveys	Interteam (MTS) transition processes and action processes (h)	Intrateam action processes, goal hierarchies, MTS performance (i)	Interteam processes predicted MTS performance (above variance explained by intrateam processes); interteam action processes maximized MTS performance when teams faced highly interdependent goal hierarchies.
Marrone, Tesluk, & Carson (2007)	31 consulting teams of MBA students	Surveys	Ambassadorial Scouting (a)	Self-efficacy, role responsibilities, role overload, team external focus, team performance (g), team viability	Team member boundary spanning behaviors were predicted by individual- and team-level variables and resulted in higher role overload for individuals. Greater team-level boundary spanning reduced overload and explained team effectiveness.

(continued)

Table 2 (continued)

Study	Team type and number	Method	Boundary spanning variables studied	Other variables studied	Key findings
Richter, West, Van Dick, & Dawson (2006)	53 work groups	Survey	Intergroup contact (c)	Group and organization identity, intergroup conflict, productivity of intergroup dyad (g)	Dual-identity model supported such that the relationship between group identification and intergroup relations is maximized when organizational identity is also high.
Tesluk & Mathieu (1999)	88 maintenance and construction road crews	Interviews, observation, surveys	Problem-management actions/strategies toward external performance barriers (b)	Crew characteristics, performance barriers, crew performance (g), cohesion, satisfaction, intent to stay	Crews that engaged more problem management encountered fewer performance barriers. Crew characteristics (teamwork, self-management, internal and external leadership) correlated with external management.

Note: For the letters in parentheses: a = measured via aggregation of individual boundary spanning behavior across team members; b = measured via team-referent scale items; c = measured via frequency of communication; d = measurement approach not provided; f = assessed via qualitative methods; g = performance assessed by team managers or other external parties; h = measured via behaviorally anchored rating scales by subject matter experts; i = performance assessed by objective ratings.

ideas; boundary spanning self-efficacy (Marrone et al., 2007) and member tenure and experience (Ancona & Caldwell, 1990) have emerged as strong motivational drivers of member boundary spanning behavior, as these factors likely to lead to higher goal setting and more realistic task performance strategies, which should promote success and continued confidence. In addition, team leader strategies toward the external environment have also been shown to provide strategic direction and support needed for assisting members in understanding their environment, managing external interactions, and recognizing the performance impacts of external team processes (Ancona, 1990). Finally, in their multilevel study of MBA student consulting teams, Marrone et al. (2007) found that, in addition to member self-efficacy, member role responsibilities and team external focus positively predicted individual boundary spanning behavior within the team.

Outcomes and mediating mechanisms (Path B). As alluded to above, challenges such as role stress, conflict, and ambiguity are associated with organizational boundary spanning roles based on early organizational studies (see Katz & Kahn, 1978). However, it is also likely that positive outcomes are realized for members who engage in boundary spanning on behalf of the collective, yet less research attention has been given to this line of inquiry. Marrone et al. (2007) found that although individual boundary spanning behavior did result in the experience of role stress, a cross-level finding revealed that through increased boundary spanning at the team level (e.g., a maximal number of boundary spanners), individual role stress was significantly mitigated. Several possible mediating mechanisms were discussed. For example, the resources gained from team boundary spanning likely help teams accomplish task expectations, thereby reducing the demands and overall stress on its individual members.

Bottom-up processes to compose team-level boundary spanning (Path C). To date, little direct attention has been given to specifying how member boundary spanning behaviors combine at the team level or to understanding the processes through which members develop shared understandings regarding their environment, how to manage it collectively, or the strategic importance of doing so. Related, despite the practical value for organizations, to my knowledge few studies to date have been conducted regarding the appropriate number of boundary spanners in a team (e.g., the use of a single focal boundary spanner vs. shared boundary spanning responsibility across all members).

Team Boundary Spanning

Much of the research stemming from Ancona and Caldwell's seminal findings focuses on relationships at the team level. Key antecedents (Path D) emergent from this work are reviewed below and include leadership, team characteristics, and composition. Analogous to the member level, it is expected that factors related to team motivation, competency, and development of appropriate task strategies will similarly predict team boundary spanning behavior.

Team leadership. Strong support exists for the role of directive, task-oriented leadership (internal as well external to the team) in supporting the team boundary activity of externally

dependent teams. Selected examples include Ancona's (1990) qualitative study of consulting teams, which demonstrated that those teams led by directive team leaders (described as offering direction and ideas, encouraging problem solving, enabling teams to take imitative, and delegating work tasks) exhibited a stronger collective external orientation and realized higher performance over time. Tesluk and Mathieu (1999) similarly found that the external supervisor behaviors of ongoing maintenance road crews (e.g., providing necessary resources, setting clear direction, and encouraging crews to set their own goals) positively correlated with crew actions and strategies directed toward the environment. Leader *coaching* (i.e., direct interventions designed to maximize team performance by supporting team processes, strategies, and/or motivation; Hackman, 2002) may also be important. In a study of operating room teams by Edmondson (2003), team leader coaching promoted team boundary spanning (i.e., efforts to coordinate schedules and objectives with others in the organization). However, Druskat and Kayes (1999) studied maintenance and production teams in a *Fortune* 500 manufacturer and found a consistent pattern of negative correlations between external leader coaching and team cross-boundary activities (although none were statistically significant). The authors speculated that coaching interventions created frustration or overdependence on the leaders.

Findings also suggest direct relationships between the team leader's own external boundary spanning actions and team outcomes (Ancona, 1990; Ancona & Caldwell, 1992a). Druskat and Wheeler's (2003) qualitative study in a manufacturing setting provides insights revealing that most successful external team leaders consistently engaged in both internally and externally focused behaviors for their teams, shifting their attention and effort back and forth between the team itself (e.g., building team trust, diagnosing team member behavior, investigating problems, delegating authority, etc.) and the broader organization in which the team operated (e.g., social and political awareness, seeking information from outside parties, obtaining external support). The impact of leader behavior on the boundary spanning behavior of the team or its members was not addressed.

Other team characteristics. The role of other internal team characteristics in supporting a team's competency and/or motivation for boundary activity has also been examined. First, Edmondson (1999) found that team psychological safety was the strongest facilitator of team boundary spanning activities, along with context support (such as adequate information, resources, and training), in her study of teams within an office furniture manufacturer. It was suggested that psychological safety enhanced the likelihood of members taking on interpersonal risks and challenges associated with externally oriented behaviors. In the opposite causal direction, boundary work facilitated both team psychological safety and performance in Faraj and Yan's (2009) study of software development teams. The authors hypothesized but did not find support for a mediating role of psychological safety between team boundary work and performance. Second, Richter, West, van Dick, and Dawson (2006) found support for group identification, organizational identification, and intergroup contact as important antecedent and moderating variables. Investigating 53 work groups in five health care organizations, results suggested that intergroup relations were maximized when groups simultaneously identified highly with their group as well as with their organization. Drawing on social identity theory, the authors note that the addition of high organizational identity allows

groups to focus on a positive superordinate identity while also maintaining their distinctiveness, thereby mitigating an otherwise negative impact of high group identity on intergroup relations.

Team composition. Finally, Ancona and Caldwell (1998, 2000) have written at least two chapters focused on how to compose teams to carry out necessary boundary spanning activities. In particular, team functional and tenure diversity are asserted and have been found to have positive direct effects (Ancona & Caldwell, 1992b; see also Joshi et al., 2009), as a broader range of expertise and knowledge enables greater and more diverse external communication; yet, it was also noted that these same attributes may have negative indirect effects by hindering key internal team processes. In addition, four aspects of team configuration have been offered—using experts, shifting composition via part-cycle members, combining full- and part-time members, using core and peripheral members—for teams striving to include external linkages without enlarging team size so greatly that internal communication is strained, efficiency is comprised, and team identity is diminished. Empirical research is limited, and to my knowledge, no work to date has directly tested these team configurations or their impacts on external team behavior.

Contextual (top down) antecedents (Path E). Ancona's (1990) early work suggests that characteristics of the broader environment matter and that open-systems theory has traditionally advocated for the role of environmental uncertainty in driving an organization's boundary spanning activity (see Katz & Kahn, 1978). Ancona and Caldwell's (1990) study of new product development teams provided support for the predictive value of resource availability (less availability promoted more external scanning for information) and for the stage of product development, which is often dictated by parameters set at the organizational level (increased ambassadorial activity occurred during creation and diffusion stages than during development). Empirical research is sparse, yet extant research has also offered some support for the role contextual variables such as resources and training (Edmondson, 1999) and organizational differences (Ancona & Caldwell, 1990) in predicting team boundary spanning activity.

Outcomes and mediating mechanisms (Path F). The positive impact of team boundary spanning on team outcomes, including team innovation, efficiency, and goal attainment, has been solidly established in both early seminal work by Ancona and subsequent advances. Gladstein's (1984) study found that external contextual factors (as opposed to internal processes) were related to sales team performance as measured by sales revenue and underscored the need for teams research to explicitly consider team boundary spanning in predicting team outcomes. Shortly following, Ancona (1990) focused on consulting team approaches toward their environment, as advocated by their leaders, and found that probing teams who had the highest external orientation were rated as the highest performers by clients. Ancona and Caldwell (1992a) similarly revealed that new product development teams exercising a comprehensive strategy regarding their environment (i.e., drawing upon various forms of external activity at different times, namely, ambassadorial and task coordination) were the highest performers and most innovative teams over time, whereas teams engaged

in low levels of all external activities or prolonged scouting suffered poor performance. More recently, Marrone et al. (2007) found that team-level boundary spanning positively predicted consulting team performance as rated by clients and team-rated viability. Interestingly, potentially not unlike varied outcomes at the individual level, existing work at the team level also provides some evidence for a tenuous relationship between team external activity and certain outcomes (e.g., probing teams also reported lower member satisfaction and team cohesion in the short term; Ancona, 1990).

Possible mechanisms may explain the benefits of team boundary spanning for improved team performance. For example, qualitative data suggest that higher levels of team boundary spanning may connect a team to more highly valued resources, such as information, feedback on progress, and support from key external parties needed for the completion of tasks (Ancona, 1990). In addition, should internal team functioning improve as a result of team boundary spanning (Choi, 2002), the resulting process benefits—such as shared understandings across members regarding external demands, higher cohesion internally, and stronger internal communication and coordination (e.g., Edmondson, 2003)—may in part explain subsequent improvements in team performance.

That stated, empirical work examining the potential interplay between external team activity and internal team functioning indicate that this relationship is complex and may vary depending on the time elapsed, the type of boundary spanning activity displayed, and environmental conditions. For instance, as noted above, Ancona's (1990) findings revealed that probing consulting teams with the highest external outreach also experienced decreases in internal team satisfaction and cohesion, although these adverse impacts were mitigated over time as the benefits gained from boundary activities in the form of additional resources, clarity of expectations, and smoother external interactions became evident. Ancona and Caldwell's (1992a) data suggest this interplay may vary by type of activity and demonstrate strong positive correlations among ambassadorial and internal team processes and cohesion, but negative correlations among general scouting for information and these same indicators of internal team dynamics. Moderating conditions may also exist; Tesluk and Mathieu (1999) found that ongoing maintenance road crews who actively managed their environment encountered fewer obstacles and under the condition of high-performance barriers were also more cohesive (interestingly, under the condition of low performance barriers, crew boundary spanning decreased cohesion).

Moderators (path not shown). Finally, surprisingly little empirical attention has been paid thus far to possible moderating conditions of the team boundary spanning–team performance relationship. Faraj and Yan (2009) were among the few researchers to test and find moderators. Studying 64 software development teams, the authors noted that relationships between boundary work and team performance were moderated by task uncertainty and resource scarcity (e.g., when facing low resource scarcity, boundary spanning increased team performance yet was counterproductive under high resource scarcity). External interdependence—or the extent to which teams depend on input from their environment or vice versa—has also been conceptually advanced as a key moderating variable strengthening the degree to which team external activity relates to team performance (see Choi, 2002) and has been empirically examined in the MTS literature (reviewed in the next section).

Bottom-up processes to form network level boundary spanning (Path G). Similar to emergent bottom-up processes by which member actions compile to form team-level boundary spanning, the mechanisms through which the boundary spanning actions of multiple teams/actors compile to construct network-level phenomenon are not yet well known. Marks et al. (2005) submitted that analogous to team interdependence, MTS goal hierarchies bring about varying degrees of cross-team interdependencies (ranging from pooled to intensive arrangements) that determine the level of interteam coordination processes. Evidence for homologous processes across the team and network levels was found with respect to interteam action processes.

Network Boundary Spanning

At this highest level in Figure 1, research directly considers the aggregated boundary spanning actions across multiple teams and/or teams and external actors comprising the network. Thus, network boundary spanning should be strongly predicted by attributes of the broader network and are expected to produce associated outcomes at this higher level. Work in MTS has primarily advanced this line of inquiry and is reviewed below.

Antecedents (Path H). Following a functional approach to leadership, DeChurch and Marks (2006) designed a training manipulation that engaged team leaders to attend to either within-team or between-team processes and found that engaging in the latter significantly affected the extent and effectiveness of interteam coordination activities exhibited across three component teams engaged in an F-22 battle simulation task. Leaders trained to attend to between-team processes assisted the teams in jointly developing strategies for coordinating work tasks and monitoring their competitive environment. Thus, similar to findings at lower levels, some supports exists that factors likely related to motivation, competency, and/or task appropriate strategies—such as leadership influence—predict network-level boundary spanning.

Outcomes, mediating mechanisms, and moderators (Path I). MTS studies have also demonstrated that the extent to which a system (or network) of teams effectively synchronizes its efforts is a strong predictor of collective success, above and beyond any single component team's internal team processes or performance (DeChurch & Marks, 2006; Marks et al., 2005). As noted above, Marks et al. also noted that interteam action processes (e.g., coordination of tasks, monitoring of activities, etc.) were most strongly predictive of performance when component teams faced highly interdependence tasks. Fieldwork has also supported the role of network boundary spanning in attaining organization goals related to innovation, productivity, and adaptation. Edmondson (2003) demonstrated that the boundary spanning activity across interdependent operating room teams played a critical role in supporting and implementing the hospital's objective of introducing new and minimally invasive cardiac surgery technology. Further, Cross et al. (2000), in a case study at National Cooperative Bank (NCB), noted that the success of the bank's restructuring of its lending process rested on multiple teams exhibiting boundary spanning within the organization (e.g., interteam consultations among credit and servicing functions to solve problems in real time)

and outside the organization (e.g., interfacing directly with customers to inform marketing and servicing decisions) and allowed NCB to successfully adapt in the face of strong external pressures in competition, changing technology, deregulation, and globalization.

The mediating processes through which network boundary spanning produces outcomes have not yet received a great deal of direct empirical attention. It is noted, however, that effective synchronization of efforts across teams enables the exchange of the inputs, processes, or outcomes of one team necessary for another team's performance and, therefore, for the attainment of collectively held goals (DeChurch & Marks, 2006; Edmonson, 2003).

Conclusion. Overall, the work on team boundary management spans multiple levels of analysis and provides a rich variety of settings, team types, and research methods. Looking across these diverse studies, team boundary management is a significant contributor to team performance outcomes, including team innovation, efficiency, and goal achievement, and higher level outcomes such as network performance and successful implementation of organization-wide change initiatives, innovation, and adaptation. These benefits appear to exist across a variety of team types, such as project/development teams (e.g., consulting, new product, software development), production/service teams (e.g., maintenance crews, manufacturing), and action teams (i.e., operating room and experimental combat teams).

Future Research Directions

Stemming from a consideration of both the current state of team boundary spanning research as well as findings from other related research areas, as described below, five broad research directions are put forth for consideration. The first three focus primarily at the team level of analysis, whereas the last two explore issues across member, team, and network levels.

Research Direction 1: How Teams Span Boundaries Most Effectively

Although previous research has recognized the importance of external team processes (Gladstein, 1984), more attention is needed to examine at a finer-grained level *how* a team can most effectively carry out critical boundary spanning processes. This research direction can take several forms, such as examining (a) how teams attend to competing internal and external demands, (b) the conditions (environmental and internal) under which external activity is more or less likely to hamper internal team dynamics, (c) who, or how many members, boundary spans, and (d) the use of virtual communication tools to facilitate boundary activities.

First, boundary spanning teams often face specific challenges surrounding the need to continually balance potentially competing demands for effective internal and external team processes (see Choi, 2002). Members must continuously make choices about how to allocate their attentional resources across various efforts (Kanfer & Ackerman, 1989), and furthermore, it has been noted that when facing either competing goals or limited resource capacity, as is often the case in organizational teams, feedback directs individual and team attention toward *one* specific goal over another (DeShon, Kozlowski, Schmidt, Milner, & Wiechmann, 2004). Following Ancona (1990), higher performing teams focused attention externally first

and then, utilizing the information and resources gained, attended internally toward the development of strategies and coordination of work tasks. Consequently, team leaders can play critical roles in helping teams effectively manage their environments by monitoring the team's task and/or stage of development and providing targeted feedback to direct attention toward either external or internal goal progress as appropriate (rather than expecting teams to attend to both simultaneously). While issues such as these concerning the allocation of resources across internal and external activities have been highlighted conceptually (e.g., Choi, 2002), direct empirical attention is sparse. Thus, future research might ask: *How do teams and members allocate attention/resources across internally and externally directed activities, and how are such decisions influenced by feedback provided by the team's leader? Further, do teams that focus externally initially become higher performers than teams that focus internally initially?*

Second, additional exploration of the conditions under which carrying out external activities is more likely to facilitate productive internal team functioning presents a primary area for future research attention. Building from the empirical results reviewed earlier by Tesluk and Mathieu (1999) and Faraj and Yan (2009), future research should explore further if engaging in boundary spanning when facing adverse environmental conditions, such as high levels of uncertainty and situational constraints, simultaneously strengthens internal team functioning, and if so, why? It may be that teams that proactively and effectively manage their environments under adverse conditions are able to unite together despite such challenges. Under more stable, predictable conditions, however, external outreach may be perceived as unnecessary and/or may prove unproductive and therefore weaken team internal dynamics. Therefore, future research questions may include: *How do environmental conditions impact the relationship between external and internal team dynamics? Do adverse conditions (e.g., uncertainty) perhaps necessitate a greater need for boundary spanning and increase the likelihood and salience of benefits to the team, thereby fostering internal cohesion and functioning? Under routine conditions, will boundary spanning be unrelated or negatively related to internal team functioning?*

Internally, the existence of team support structures should further help teams that boundary span to sustain positive internal team dynamics. In particular, integrations of social network perspectives as well as early findings from the organizational boundary spanning literature help to further inform this issue. For example, because boundary spanning not only involves the team's efforts to engage the external environment but also requires the transmission of resulting external information and knowledge back into the team itself (e.g., Tushman, 1977), supportive internal team network structures are likely to be essential for effectively disseminating information obtained through boundary spanning to the team's members. Specifically, the development of a dense internal information network, reflective of numerous informational ties among all team members (Sparrowe, Liden, Wayne, & Kraimer, 2001), should enable the information and other resources obtained from external sources to more efficiently and effectively reach all team members as they work toward task completion. In contrast, for these same reasons, high network centrality (when network ties are concentrated in just a few members) could impede the team's collective ability to identify, disseminate, and maximize externally derived resources. This is particularly true for teams facing high levels of internal interdependence as communication flow is concentrated

in only a subset of members. Tie strength may additionally be important when information gained externally is highly complex or tacit (Hansen, 1999). Thus, research questions for future study might ask: *To what extent do internal network structures assist or hinder the impact of team boundary spanning actions on internal team functioning and performance? Is boundary spanning activity more strongly related to team processes and performance in teams exhibiting high network density and less (or negatively) related to team performance in teams exhibiting high network centrality?*

Yet another issue involving “how” to carry out boundary spanning activities involves *who* (e.g., team members, team leaders, both) should be carrying out boundary spanning responsibilities. Ancona and colleagues’ early work depicts team leader approaches to boundary management, which are then shown to affect the team’s overall external orientation and level of team boundary spanning activity, yet there is little discussion as to who is or who should be carrying out these critical activities within the team. Thus, important questions arise for future study: *Is this a job or role best reserved for only a single team member or leader, or given its importance and scope, should boundary spanning activity be the responsibility of at least a critical mass, if not all, team members? Under what conditions does this answer depend?* An integration of research in organizational boundary spanning research by Tushman (1977) may offer some insight here by noting that the ideal composition of boundary spanning within R&D units varied depending upon the task at hand. When facing more complex and less routine tasks, high-performing units consistently relied on an increased number of boundary spanning roles, whereas low-performing units utilized a centralized structure relying on a single person, the supervisor, to carry out boundary spanning activities. However, the opposite pattern was found for successful projects facing environments characterized by uncertainty (e.g., unexpected changes or turbulence) or high external demands (e.g., numerous and diverse). Open questions such as these call for more direct attention to bottom-up processes of emergence as well as key factors influencing the efficacy of one type of composition over another.

Finally, as Kirkman and Mathieu (2005) note, teams vary in the extent of their virtuality, defined as the extent of reliance on virtual tools, the informational value provided by the tools, and the synchronicity of virtual interaction across members. They predict that because boundary spanning activities connect dispersed others (located in external teams, organizations, and even countries), the requirements for and frequency of use of electronic communication tools (vs. face-to-face) significantly increase. These factors also relate to the use of less rich and asynchronous virtual interactions. Broadly, one research question might be: *How does the use of virtual tools affect engagement in boundary spanning activities, including the frequency, quality, effectiveness, and efficiency of external and internal communications?*

Research Direction 2: When Teams Span Boundaries

Although issues surrounding time have been raised in relatively few studies of team boundary spanning, integrations of emergent findings there and in the broader teams literature provide strong cause for further exploration in future research. Key questions to consider involve the impact of product life cycles, team development stages, and time duration on boundary spanning activity and effectiveness.

First, at least some existing data suggest that product life cycles are critical antecedents to the type and level of team boundary spanning behavior. Ancona and Caldwell's (1990) study of new product development teams revealed that ambassadorial activities were highest during the early creation and late diffusion stages of product development and lowest during the middle stage of development, but there have been no attempts yet to replicate. Also, although impending product-related deadlines appeared to decrease boundary spanning activity (Ancona & Caldwell, 1988), Choi (2002) theorizes that deadlines heighten needs for external information and feedback and therefore call for increased external activity. Following Choi's assertions, related teams research suggests that the attributes of the team's temporal environment (organizational as well as intergroup), including deadlines and time pressures, influence decisions about how to spend time and affect the importance and weighting of outcome criteria (Mohammed, Hamilton, & Lim, 2009). Thus, further work might explore the following: *How do teams respond via boundary spanning actions to events in their temporal environment that signal the need for external outreach and/or make more salient externally imposed deadlines? Under what conditions might these same temporal markers create undue pressures and/or signal threats to the team that in turn decrease external outreach?*

Turning to the team's own internal life cycle, Gersick's (1988) seminal work on group development offers compelling evidence for predictable variation in the overall level of boundary spanning activity over the team's life. Her findings depicted a "punctuated equilibrium" whereby teams unanimously experienced major upheaval at their midpoint transitions (halfway between first meeting and official deadline) that involved questioning old patterns, making changes to approaches toward work, and seeking advice from external supervisors. In contrast, during periods prior to and after the midpoint, teams experienced strong inertia and remained largely insulated from their external environments. Even finer grained, Joshi et al. (2009) draw on Marks et al.'s (2001) episodic framework to assert that the team's task cycle may predict the particular *type* of boundary activity displayed. Future research aimed directly at developing theory and testing these finer grained assertions is needed. Most broadly: *Do internal team cycles affect the overall level and type of boundary spanning activity such that external outreach is lowest before and after midpoint transitions and highest around initial meetings, midpoint transitions, and last meetings?*

Yet another issue considers if the performance benefits associated with boundary spanning activity differ in meaningful ways over time. As one example, Ancona and Caldwell (1992a) found that although ambassadorial activities significantly predicted team operations and efficiency at Time 1 (approximately midway at 10 months), they did not significantly predict similar performance measures at Time 2 (during final stages 2 years later). Task coordination activities held the opposite pattern with respect to team innovation across the two time periods, whereas prolonged scouting for information was found to negatively affect performance at Time 1 and Time 2. Future work might directly examine: *How does the passage of time serve to enhance some of a team's boundary spanning efforts (e.g., coordination of task performance actions) while hindering the efficacy of other efforts (e.g., general information search actions)?*

Finally, for teams that are highly dependent on their external environments, developing shared and accurate understandings of the temporal rhythms and priorities of key external groups is likely critical for carrying out boundary spanning efforts in ways that maximize team performance (e.g., Bluedorn & Standifer, 2004). In addition, it would be interesting to

explore if holding a long-term time orientation enables teams to more readily perceive strategic value in external connections and, thus, more likely to pursue and persist in such activities despite the necessary efforts and resources associated with doing so. Some related support has emerged suggesting that differences in team member temporal personalities and perspectives (future vs. present) and urgency (time urgent vs. not) may combine in specific ways that either help or hinder team behavior and resulting performance (cf., Mohammed et al., 2009). Purposeful and direct examinations of time-related factors such as these can significantly extend our understanding of the temporal complexities inherent to team boundary spanning activity and its influence on team performance.

Research Direction 3: Contingencies to the Boundary Spanning–Performance Relationship

It is both notable and surprising that little attention has been given to exploring contingencies to the team boundary spanning–team performance relationship, especially given the strong assertions from early work that boundary spanning may be most critical for certain team types under certain conditions (e.g., highly externally dependent teams facing uncertain environments; Ancona, 1990).

Future research may also explore if structural elements, such as team size, serve as meaningful influences on a team's ability to carry out boundary spanning activity more or less effectively. For instance, because smaller teams may find it easy to meet frequently and communicate often, they may be less likely to become “underbounded” by boundary spanning activity and better able than larger teams to leverage and utilize the benefits gained from boundary spanning (e.g., smaller size allows for more efficient dissemination of external information to all internal members). Despite the significant practical implications such explorations may have for organizations employing boundary spanning teams, to my knowledge no study to date has examined the potential impacts of team size on boundary spanning effectiveness. Another structural element—team diversity—can also be explored further, following Cummings (2004), who offers related empirical support suggesting that the relationship between group performance and external knowledge sharing is maximized when the group is also highly diverse. Consequently, one line of inquiry for future research is: *How might team structural elements (such as size and diversity) maximize the team boundary spanning–performance relationship?*

Future research might also explore external interdependence—or interdependences across multiple teams, or across teams and external actors—as a key moderating variable in predicting both team- and network-level performance outcomes. A robust finding in the team literature asserts that the level of interdependency internally influences how strongly internal team processes predict team performance (LePine, Piccolo, Jackson, Mathieu, & Saul, 2008). In a homologous manner, the level of external interdependence across teams should influence how strongly boundary spanning actions across teams predict their collective performance (Marks et al., 2005). Similarly, the level of external dependence experienced by a single boundary spanning team is asserted as a fundamental factor heightening the need for and efficacy of team boundary activities (Ancona, 1990; Choi, 2002). Thus: *To what extent*

does external interdependence moderate the boundary spanning–performance relationship? Is the nature of this relationship homologous to findings related to internal interdependence at the team level?

Research Direction 4: Additional Antecedents at Member and Network Levels

Given the strong performance benefits associated with team boundary spanning and coupled with an understanding that such actions can require considerable effort and resources on the part of teams and team members, it is important for future research to continue exploring factors that promote (or hinder) engagement in boundary spanning behavior. First, through integrations of related research in the organizational boundary spanning and marketing domains, additional insights into potential drivers of member boundary spanning in a team context emerge.

Team member roles and beyond. While team-level boundary management has been the focus of more recent attention, the general topic of boundary spanning has a rich conceptual and empirical history within the social psychology domain, focusing primarily on individuals occupying boundary spanning roles within their organizations. This work, which includes the work of Katz and Kahn (1978), Tushman (1977), and others, explores the nature of organizational boundary spanning roles, facilitators and constraints on role behavior, and anticipated outcomes associated with such roles for individuals and their organizations. For example, Katz and Kahn and others provide strong evidence for the persuasive influence of roles in directing and facilitating members to engage in boundary spanning. While the delineation of team member roles has previously been suggested as critical for accomplishing work and enhancing team effectiveness (see the review by Mathieu et al., 2008), examinations regarding the development and implementation of team member roles for executing the team boundary spanning function are sparse. Further examination offers the potential for a deeper understanding of the nature and influence of boundary spanning roles signaled to, adopted by, and carried out by team members throughout their projects.

A second domain of related research lies within the marketing literature, which has studied salespersons (e.g., Singh, 1998) and customer-service employees (e.g., Bettencourt et al., 2005) as critical boundary spanners. From this work, team boundary spanning researchers may explore further antecedents, such as trust and fairness, and outcomes, such as job satisfaction, commitment, and role stressors, that have been underscored in the marketing literature. As examples, job satisfaction and organizational commitment significantly reduced role stress experienced by organizational boundary spanners in sales (e.g., Singh, 1998) and, in addition to perceptions of justice, directly increased boundary spanning behaviors by front-line customer service employees (e.g., Bettencourt & Brown, 2003; Bettencourt et al., 2005). If applied to the team context, future research may explore if such factors increase member motivation to engage in and persist with critical boundary spanning efforts for their teams.

Following the above, a promising line of inquiry for future research can ask: *How does integration with related research areas further explain member engagement in critical boundary spanning behaviors on behalf of their teams? Might individual characteristics such as team member roles, job satisfaction, organizational commitment, and justice*

perceptions positively affect motivation for and engagement in boundary spanning by individual team members?

Network-level properties. Finally, a consideration of the properties attributed to network-level variables such as network density and centralization holds strong promise for both characterizing and predicting interrelationships across component teams and actors in the boundary spanning network. Other variables, including network-level leadership (DeChurch & Marks, 2006), organizational and cross-organization goals (Marks et al., 2001), and network composition (analogous to composition at the team level) are worthy of future study at this level.

Research Direction 5: Additional Outcomes at Member and Network Levels

Finally, future research on team boundary spanning should continue to move beyond studying impacts on team performance to exploring additional outcomes for members and their embedding networks (be these within or across organizations). Although not exhaustive, several avenues for future exploration emerge from integration with related research.

Team member reputation and leadership. An interesting story emerges when integrating research on organizational boundary spanning with research from a social network perspective. Although the former suggests adverse consequences related to boundary spanning roles, such as role conflict, ambiguity, and overload (e.g., Katz & Kahn, 1978), the latter suggests that engaging in boundary spanning for one's team may also result in substantial positive gains, such as enhanced influence, reputation, and ability to make valued contributions. Because team members who span boundaries establish connections to key external parties, such as clients, upper management, and other professionals, they also occupy structurally advantageous positions. According to social network theory, these positions, often referred to as "structural holes" or "structural bridges," can provide unique sources of power and influence to the boundary spanners as critical resources flow from external sources through boundary spanners to the team (Burt, 1992; Pfeffer & Salancik, 1978); therefore, boundary spanners may routinely be perceived as highly influential leaders by their teammates.

Furthermore, to the extent that boundary spanners are able to simultaneously occupy highly central positions in the internal team networks, these positive gains should be maximized as centrality allows for more efficient transfer of knowledge and resources obtained from external sources to the team, therefore enabling the contributions and efforts by the boundary spanner to be more readily recognized by other team members (see Brass, 1984). Because achieving such positive gains may increase a team member's subsequent engagement in critical boundary spanning behaviors, future research questions exploring how, why, and under what conditions these positive outcomes are maximized are important. Example research questions thus include: *Does member boundary spanning behavior positively affect one's reputation and leadership as perceived by teammates? Are these perceptions strongest when the boundary spanning individuals are also highly central within the team's internal networks?*

Organizational outcomes and beyond. Finally, I encourage future research to continue to explore higher level outcomes. At the organizational level, given that team efforts to span

their boundaries result in the coordination of complex tasks, knowledge, and resources across organizational functions needed for organizational innovation (Hargadon, 1998), learning (Edmondson, 1999), and customer satisfaction (Bettencourt et al., 2005), future studies should seek to directly incorporate these organizational-level variables. Furthermore, through synchronization of efforts across interdependent units (rather than independent or sequentially independent activities), network boundary spanning can also be linked to additional variables such as time compression or efficiency in operations (Mohrman et al., 1995).

Finally, network-level boundary spanning processes are extremely relevant to a number of disciplines outside of the management domain, including areas such as public health administration (e.g., Merrill, Bakken, Rockoff, Gebbie, & Carley, 2006), software development (e.g., Cataldo, Herbsleb, & Carley, 2008), and even counterterrorism (e.g., Moon & Carley, 2007). These areas are concurrently using complex modeling and simulation techniques to better understand interdependencies, coordination requirements, and communications existing within dynamic and often geographically dispersed networks of actors, knowledge, information, and tools. Throughout this work, challenges associated with managing such interdependencies are highlighted, such as poor understanding or management of the social and informal dynamics underlying information networks in public health agencies (Merrill et al., 2006) and the confusion or lack of awareness surrounding the strong performance detriments caused by inadequate communication across members and teams in software development companies (Moon & Carley, 2007). Therefore, advancement in the management discipline might significantly help organizations in a variety of disciplines to identify and more effectively manage the factors that facilitate, hinder, and maximize boundary spanning actions.

Measurement Issues

A few final comments regarding measurement are important given the potential impact on all five future research directions. As explained in the note to Table 2, several measurement approaches have emerged within the existing literature. Previous work has often utilized qualitative methods to assess the nature and extent of external team processes in their samples of teams (e.g., Ancona, 1990; Edmonson, 1999). Quantitatively, some researchers have captured team boundary spanning through scale items assessing team member boundary spanning actions and then aggregating to the team level of analysis (e.g., Ancona & Caldwell, 1992a; Marrone et al., 2007). These items come from Ancona and Caldwell (1990, 1992a), who rigorously developed and validated a 21-item scale tapping three distinct factors of team member boundary spanning behavior: ambassadorial, task coordination, and scouting. Alternative measurement strategies have included the use of team-referent scales (e.g., Druskat & Kayes, 1999; Faraj & Yan, 2009), whereas others have utilized various social network techniques capturing the frequency and quality of external contact (e.g., Cummings, 2004; Hansen, 1999), which can be particularly useful for examining patterns of member-, team-, or network-level boundary spanning. Ancona and Caldwell (1992a) noted that assessments of the nature or pattern of external activity were notably superior to measures that capture merely the frequency of external communication.

Additional steps in the area of team boundary measurement can help to further our understanding of this important team phenomenon, discern how best to measure it, and enhance the

robustness of empirical findings. First, because the form and nature of emergent processes underlying team boundary spanning are not likely universal, but rather vary depending upon particular team and task requirements, future work should strive to provide even greater theoretical clarity and specificity regarding these underlying assumptions and how they align with the selected measurement approach. This will help to guide subsequent measurement decisions in follow-up work, allow for more informative comparisons across studies where different measurement approaches are used, and promote a greater understanding of the complex and varied ways teams can and do carry out the boundary spanning function. Second, where possible and when appropriate, it is recommended that future studies draw on Ancona and Caldwell's (1992a) scale, as it is established and provides comprehensive coverage of the primary boundary spanning activities. For researchers seeking even finer grained analyses, the measurement of team boundary spanning activity might be delineated further through explicit consideration of the target itself. New items or scales may be developed to capture idiosyncratic behaviors most relevant for spanning to particular targets (e.g., representing the team may involve additional or unique actions when interacting with specific parties outside vs. inside the organizational boundary; additional items may be needed to more fully capture synchronization of work processes across mutually interdependent teams). Moving to a consideration of more general measures, to my knowledge few attempts have been made to create a measure tapping a team's overall external orientation or propensity to engage its environment. Future development of a broad, global measure may offer an efficient way to capture overall team boundary spanning intention or activity and may inform the eventual development of other related constructs, such as team proactivity or team initiative.

Finally, the measurement of team boundary spanning in future work is likely to require even greater transparency with respect to defining the team's boundary itself. The growing presence of virtual teams, ad hoc collaborations (e.g., open-source collaborations, communities of practice), and frequent organizational restructurings often result in greater-than-before input on team decisions, fluid access to team information, and frequently changing membership, therefore making it increasingly difficult for researchers (and perhaps even the participating members themselves) to discern the team's boundary. Consequently, the measurement of team spanning activity occurring across that boundary is also complicated. Applying suggestions from the social network literature (see Wasserman & Faust, 1994), to manage such complexities researchers may draw upon existing theory (e.g., definitional elements of a team, such as mutually interdependent team members, working toward a common goal, seen by themselves and others as a social identity; Guzzo & Dickson, 1996), or develop new theoretical assumptions appropriate to their particular research questions (e.g., frequency of information exchange), to determine the team's boundary. Alternatively, researchers may rely on the actors themselves (the participating members) to acknowledge and report on those they perceive to be inclusive of the collective, and in doing so, the members define the team's boundary. As changing technology and work environments continue to make possible new and creative forms of collaboration, researchers studying team boundary spanning must take even greater care to understand the complexities involved in defining the team boundary and, ultimately, in measuring team boundary spanning.

Conclusion

In summary, this review article seeks to further our knowledge of team boundary spanning through reviewing the stream of team boundary spanning literature and integrating this work with

other related research areas. Early seminal work by Ancona and subsequent empirical and conceptual advances were reviewed to devise a taxonomy of team boundary spanning actions and present extant findings across three levels of analysis: member, team, and network. Next, through integrations of this literature with related studies in organizational boundary spanning, social network theory, marketing, and other work, five primary directions for future research were explored in detail. Through this review and integration, this article seeks not only to advance team boundary research and related fields but also to benefit organizations seeking to maximize their efficiency, learning, innovation, and adaptability in today's changing business environment.

Note

1. The term *network boundary spanning* is used to reflect boundary spanning behaviors across multiple teams or team and actors comprising the network (direct exploration of a network's efforts to span to parties external to the network itself is outside the scope of this article).

References

- Ancona, D. G. 1990. Outward bound: Strategies for team survival in an organization. *Academy of Management Journal*, 33: 334-365.
- Ancona, D. G., & Caldwell, D. F. 1988. Beyond task and maintenance. Defining external functions in groups. *Group and Organization Studies*, 13: 468-494.
- Ancona, D. G., & Caldwell, D. 1990. Beyond boundary spanning: Managing external dependence in product development teams. *Journal of High Technology Management Research*, 1: 119-135.
- Ancona, D. G., & Caldwell, D. F. 1992a. Bridging the boundary: External activity and performance in organizational teams. *Administrative Science Quarterly*, 37: 634-665.
- Ancona, D. G., & Caldwell, D. 1992b. Demography and design: Predictors of new product team performance. *Organization Science*, 3: 321-341.
- Ancona, D. G., & Caldwell, D. F. 1998. Rethinking team composition from the outside in. In D. H. Grunfield (Ed.), *Research on managing groups and teams: Composition*: 21-37. Stamford, CT: JAI.
- Ancona, D. G., & Caldwell, D. 2000. Compose teams to assure successful boundary activity. In E. A. Locke (Ed.), *The Blackwell handbook of principles of organizational behavior*: 199-201. Oxford, UK: Blackwell.
- Argote, L., McEvily, B., & Reagans, R. 2003. Managing knowledge in organizations: An integrative framework and review of emerging themes. *Management Science*, 49: 571-582.
- Bettencourt, L. A., & Brown, S. W. 2003. Role stressors and customer-oriented boundary-spanning behaviors in service organizations. *Journal of the Academy of Marketing Science*, 31: 394-408.
- Bettencourt, L. A., Brown, S. W., & MacKenzie, S. B. 2005. Customer-oriented boundary spanning-behaviors: Test of a social exchange model of antecedents. *Journal of Retailing*, 81: 141-157.
- Bluedorn, A. C., & Standifer, R. L. 2004. Groups, boundary spanning, and the temporal imagination. *Research on Managing Groups and Teams*, 6: 159-182.
- Burt, R. S. 1992. *Structural holes: The social structure of competition*. Cambridge, MA: Harvard University Press.
- Brass, D. J. 1984. Being in the right place: A structural analysis of individual influence in an organization. *Administrative Science Quarterly*, 29: 518-539.
- Cataldo, M., Herbsleb, J., & Carley, K. 2008. Socio-technical congruence: A framework for assessing the impact of technical and work dependencies on software development productivity. In *Proceedings of the 2nd International Symposium on Empirical Software Engineering and Measurement*, Kaiserslautern, Germany.
- Chan, D. 1998. Functional relations among constructs in the same content domain at different levels of analysis: A typology of composition models. *Journal of Applied Psychology*, 83: 234-246.
- Choi, J. N. 2002. External activities and team effectiveness: Review and theoretical development. *Small Group Research*, 33: 181-208.

- Cummings, J. N. 2004. Work groups, structural diversity, and knowledge sharing in a global organization. *Management Science*, 50: 352-364.
- Cross, R. L., Yan, A., & Louis, M. R. 2000. Boundary activities in "boundaryless" organizations: A case study of a transformation to a team-based structure. *Human Relations*, 53: 841-868.
- DeChurch, L. A., & Marks, M. A. 2006. Leadership in multiteam systems. *Journal of Applied Psychology*, 91: 311-329.
- DeChurch, L. A., & Mathieu, J. E. 2009. Thinking in terms of multiteam systems. In E. Salas, G. F. Goodwin, & C. S. Burke (Eds.), *Team effectiveness in complex organizations: Cross-disciplinary perspectives and approaches*: 267-292. New York: Taylor & Francis.
- Denison, D. R., Hart, S. L., & Kahn, J. A. 1996. From chimneys to cross-functional teams: Developing and validating a diagnostic model. *Academy of Management Journal*, 39: 1005-1023.
- DeShon, R. P., Kozlowski, S. W. J., Schmidt, A. M., Milner, K. R., & Wiechmann, D. 2004. A multiple-goal, multilevel model of feedback effects on the regulation of individual and team performance. *Journal of Applied Psychology*, 89: 1035-1056.
- Druskat, V. U., & Kayes, D. C. 1999. The antecedents of team competence: Toward a fine-grained model of self-managing team effectiveness. *Research on Managing Groups and Teams*, 3: 201-231.
- Druskat, V. U., & Wheeler, J. V. 2003. Managing from the boundary: The effective leadership of self-managing work teams. *Academy of Management Journal*, 46: 435-457.
- Edmondson, A. 1999. A safe harbor: Social psychological conditions enabling boundary spanning in work teams. *Research on Managing Groups and Teams*, 2: 179-199.
- Edmondson, A. 2003. Speaking up in the operating room: How team leaders promote learning in interdisciplinary action teams. *Journal of Management Studies*, 40: 1420-1452.
- Faraj, S., & Yan, A. 2009. Boundary work in knowledge teams. *Journal of Applied Psychology*, 94: 604-617.
- Gersick, C. J. G. 1988. Time and transition in work teams: Toward a new model of group development. *Academy of Management Journal*, 31: 9-41.
- Gladstein, D. L. 1984. Groups in context: A model of task group effectiveness. *Administrative Science Quarterly*, 29: 499-517.
- Guzzo, R. A., & Dickson, M. W. 1996. Teams in organizations: Recent research on performance and effectiveness. *Annual Review of Psychology*, 47: 307-338.
- Hackman, J. R. 2002. *Leading teams: Setting the stage for great performances*. Boston: Harvard Business School Press.
- Hansen, M. T. 1999. The search-transfer problem: The role of weak ties in sharing knowledge across organizational subunits. *Administrative Science Quarterly*, 44: 82-111.
- Hargadon, A. B. 1998. Firms as knowledge brokers: Lessons in pursuing continuous innovation. *California Management Review*, 40: 209-227.
- Joshi, A., Pandey, N., & Han, H. (2009). Bracketing team boundary spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior*, 30: 731-759.
- Kanfer, R., & Ackerman, P. L. 1989. Motivation and cognitive abilities: An integrative/aptitude-treatment interaction approach to skill acquisition. *Journal of Applied Psychology*, 74: 657-690.
- Katz, D., & Kahn, R. L. 1978. *The social psychology of organizations*. New York: John Wiley.
- Kirkman, B., & Mathieu, J. 2005. The dimensions and antecedents of team virtuality. *Journal of Management*, 31: 700-718.
- Kozlowski, S. W. J. & Klein, K. J. 1999. A multi-level approach to theory and research in organizations: Contextual, temporal, and emergent processes. In K. J. Klein & S. W. J. Kozlowski (Eds.), *Multilevel theory, research, and methods in organizations: Foundations, extensions, and new directions*: 3-90. San Francisco: Jossey-Bass.
- LePine, J. A., Piccolo, R. F., Jackson, C. L., Mathieu, J. E., & Saul, J. R. 2008. A meta-analysis of teamwork processes: Tests of a multidimensional model and relationships with team effectiveness criteria. *Personnel Psychology*, 61: 273-307.
- Marks, M. A., De Church, L. A., Mathieu, J. E., Panzer, F. J. & Alonso, A. 2005. Teamwork in multiteam systems. *Journal of Applied Psychology*, 90: 964-971.
- Marks, M. A., Mathieu, J. E., & Zaccaro, S. J. 2001. A temporally based framework and taxonomy of team processes. *Academy of Management Review*, 26: 356-376.
- Marrone, J. A., Tesluk, P. E., & Carson, J. B. 2007. A multi-level investigation of antecedents and consequences of team member boundary spanning behavior. *Academy of Management Journal*, 50: 1423-1439.

- Mathieu, J. E., Marks, M. A., & Zaccaro, S. J. 2001. Multiteam systems. In N. Anderson, D. S. Ones, H. K. Sinangil, & C. Viswesvaran (Eds.), *Organizational psychology: Vol. 2. Handbook of industrial, work and organizational psychology*: 289-313. London: Sage.
- Mathieu, J. E., Maynard, M. T., Rapp, T., & Gilson, L. 2008. Team effectiveness 1997-2007: A review of recent advancements and a glimpse into the future. *Journal of Management*, 34: 410-476.
- Merrill, J., Bakken, S., Rockoff, M., Gebbie, K., & Carley, K. 2006. Description of a method to support public health information management: Organizational network analysis. *Journal of Biomedical Informatics*, 40: 422-428.
- Mohammed, S., Hamilton, K., & Lim, A. 2009. The incorporation of time in team research: Past, current, and future. In E. Salas, G. F. Goodwin, & C. S. Burke (Eds.), *Team effectiveness in complex organizations: Cross-disciplinary perspectives and approaches*: 321-350. New York: Taylor & Francis.
- Moon, I., & Carley, K., 2007. Modeling and simulation of terrorist networks in social and geospatial dimensions. *IEEE Intelligent Systems*, 22: 40-49.
- Mohrman, S. A., Cohen, S. G., & Mohrman, A. M. 1995. *Designing team-based organizations: New forms for knowledge work*. San Francisco: Jossey-Bass.
- Pfeffer, J., & Salancik, G. 1978. *The external control of organizations*. New York: Harper & Row.
- Richter, A. W., West, M. A., van Dick, R., & Dawson, J. F. 2006. Boundary spanners' identification, intergroup contact, and effective intergroup relations. *Academy of Management Journal*, 49: 1252-1269.
- Singh, J. 1998. Striking a balance in boundary-spanning positions: An investigation of some unconventional influences on role stressors and job characteristics on job outcomes of salespeople. *Journal of Marketing*, 62: 69-86.
- Sparrowe, R. T., Liden, R. C., Wayne, S. J., & Kraimer, M. L. 2001. Social networks and the performance of individuals and groups. *Academy of Management Journal*, 44: 316-325.
- Tesluk, P. E., & Mathieu, J. E. 1999. Overcoming roadblocks to effectiveness: Incorporating management of performance barriers into models of work group effectiveness. *Journal of Applied Psychology*, 84: 200-217.
- Tushman, M. L. 1977. Special boundary roles in the innovation process. *Administrative Science Quarterly*, 22: 587-605.
- Wasserman, S. & Faust, K. 1994. *Social network analysis: Methods and applications*. Cambridge, UK: Cambridge University Press.